

Ati Rostami

Research Data Scientist with a PhD in applied mathematics, specialising in mathematical modelling, simulation and Bayesian inference. Experienced in analysing plant growth models, building machine learning models in Python, and communicating findings through research and teaching.

Professional Experience

Present
|
2026

AI Resident

Apziva

 Remote

- Built predictive models in Python to analyse customer satisfaction and identify customers likely to subscribe to a term deposit.
- Compared logistic regression, support vector machines, random forests and gradient boosting using scikit-learn to evaluate predictive performance.
- Achieved an F1 score of approximately 0.71 for term-deposit subscription prediction using gradient boosting.

2026
|
2021

Research Data Scientist

Plant Success Centre

 Brisbane, Australia

- Developed and analysed whole-plant carbon-nitrogen models using numerical simulation to investigate shoot and root growth.
- Conducted scenario analysis and parameter sweeps to identify unrealistic model behaviour; improved the model structure by adding litter terms.
- Applied Bayesian sequential Monte Carlo to synthetic dry-mass data for parameter estimation, uncertainty analysis and assessment of parameter correlations.
- Used principal component analysis of posterior samples to assess parameter identifiability and determine which parameter combinations were informative.
- Compared measurement scenarios to understand data limitations, draw insights and provide recommendations for experimental design; communicated findings to interdisciplinary researchers.

Teaching

Present
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2023

Teaching Associate (Mathematics, Programming & Data Science)

Queensland University of Technology

 Brisbane, Australia

- Delivered workshops, practical labs and tutorials in mathematics, data science and programming to hundreds of undergraduate students across multiple units.
- Taught R in Advanced Visualisation and Data Science, supporting data analysis, visualisation and lab portfolios; also taught MATLAB, calculus, statistics and algebra, and assessed coursework.

Education

2026

PhD, Applied Mathematics

Queensland University of Technology

 Brisbane, Australia

- Thesis: Re-engineering Enduring Mathematical Models Using Novel Techniques and New Data

2015

MPhil, Pure Mathematics

K. N. Toosi University of Technology

 Tehran, Iran

- Thesis: Hyperconvex Metric Spaces

2011

BSc, Pure Mathematics

Shahrood University of Technology

 Shahrood, Iran

Contact

 Brisbane, Australia

 github.com/atefero

 [linkedin.com/in/ati-rostami/](https://www.linkedin.com/in/ati-rostami/)

Technology skills

PROGRAMMING

Python · R · MATLAB · SQL (T-SQL)

PYTHON LIBRARIES

pandas · NumPy · SciPy · scikit-learn · matplotlib

STATISTICAL & PREDICTIVE MODELLING

Bayesian inference · Classification · Logistic regression · Gradient boosting · Machine learning

ANALYSIS & VALIDATION

Scenario analysis · Model validation · Parameter estimation · Uncertainty analysis · Data visualisation

TOOLS

Power BI · Excel · Git · GitHub

Interpersonal skills

PROFESSIONAL STRENGTHS

Technical communication · Workshop facilitation · Collaboration · Problem solving · Attention to detail